

HOT FORMED HOLLOW SECTIONS

(excerpts from BRITISH STEEL TUBES)

The size range of Structural Hollow Sections manufactured by British Steel Tubes, is in accordance with BS 4848 Hot Rolled Structural Steel Sections: Part 2, Hollow Sections.

The Structural Hollow Section size range published in BS 4848: Part 2 relates generally to ISO 657 Hot Rolled Steel Sections: Part 14, Hot Finished Structural Hollow Sections, Dimensional and Sectional Properties.

STEEL

The steel is made by the basic oxygen or electric furnace process in accordance with BS 4360. For SHS there are three basic grades, 43, 50 and 55, each with sub-grades giving improved impact properties. For economic reasons, production has been standardised on grades 43C and 50C. Other grades and sub-grades can be supplied against mill orders and subject to minimum mill quantities.

CHEMICAL COMPOSITION AND MECHANICAL PROPERTIES

The ladle analyses and mechanical properties, in accordance with BS 4360 for hot finished sections are as follows:

	Grade 43C	Grade 50C		Grade 43C	Grade 50C
	%max	% max			
Carbon	0.21	0.20	Tensile Strength N/mm ²	430/580	490/640
Silicon	0.50	0.50	Yield Strength min. N/mm ²	275 [†]	355 [†]
Manganese	1.30	1.50	Min. Elongation on gauge		
Sulphur	0.05	0.045	length $5.65\sqrt{S_0}$	22%	20%
Phosphorus	0.05	0.045	Charpy V-Notch Impact Value		
Niobium	-	0.003/0.10	10 x 10mm specimen at		
Vanadium	-	0.003/0.15	0°C. Min. Ave Energy Joules	27*	27

[†] up to and including 16mm thick

* only by special arrangement

OVERSEAS SPECIFICATIONS

Other specifications, comparable to BS 4360 can be supplied; those most frequently used are as follows:

Ordinary Yield	High Yield
DIN 17100 St 44-2	DIN 17100 St 52-3
Euronorm 25 - 72 Fe 430C	Euronorm 25 - 72 Fe 510C
NBN A21-101 AE 255C	NBN A21-101 AE 355C
Onorm M3116 St 430C	Onorm M3116 St 510C
SS 14 14 12	SS 14 21 34
ASTM A36	UNI 7806 Fe 510C
UNI 7806 Fe 430C	

Other sub-grades are available

TENSILE TEST

The tensile strength, yield strength and elongation are determined from standard test pieces cut longitudinally from the sections, excluding the weld areas and away from the corners. The test pieces and testing comply with BS 18. Method for Tensile Testing of Metals.